Mechanism Questions #4

Barry Linkletter

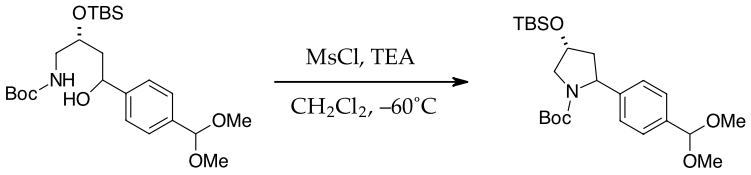
This document was produced using the LaTeX typesetting language with the Tufte-handout document class. Chemical diagrams were prepared in *ChemDoodle*.

Explain the reaction mechanisms for the transformations below. Draw out each step and show electron movement using curved arrows. Identify the likely rate-determining step and identify the driving force for each reaction.

Each reaction is a single step in a multistep synthesis described in the referenced literature. For extra experience, access each paper and describe every step in the synthesis that is presented.

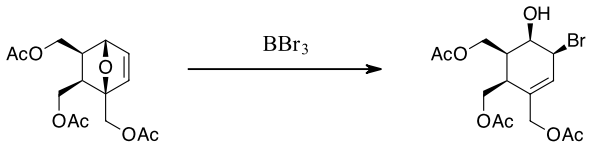
This is our final organic review set and so it is large. Keep working on these as we move forward to other topics. You can never practice too much.

1.



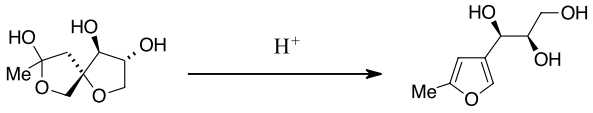
"Stereoselective Synthesis of a Broad Spectrum 1β-Methylcarbapenem, J-114,870." H. Imamura et al., *Tetrahedron*, **2000**, *56*, 7705-7713. https://doi.org/10.1016/S0040-4020(00)00685-2

2.



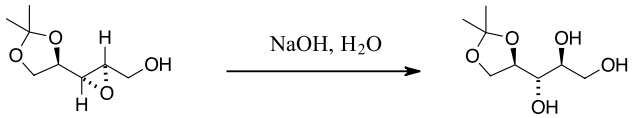
"Synthesis of a 5-Hydroxypiperidine-2,3,4,6-tetramethanol, a New 2,6-Dideoxy-2,6-iminoheptitol Derivative." N. Jotterand, P. Vogel, *J. Org. Chem.*, **1999**, *64*, 8973-8975. https://doi.org/10.1021/jo991228b

3.



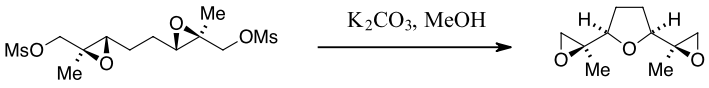
"Total Syntheses of Sphydrofuran, Secosyrins, and Syributins." P. Yu et al., *J. Org. Chem.*, **1997**, *62*, 6359-6366. https://doi.org/10.1021/jo970476+

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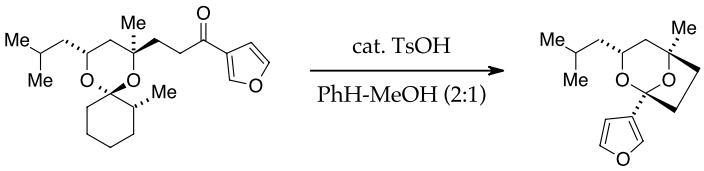
"Total Synthesis of (−)-Mniopetal E, a Novel Biologically Intriguing Drimane Sesquiterpenoid." Y.R. Nishimaki et al. *J. Org. Chem.*, **2000**, *65*, 8595-8607. https://doi.org/10.1021/jo001004p

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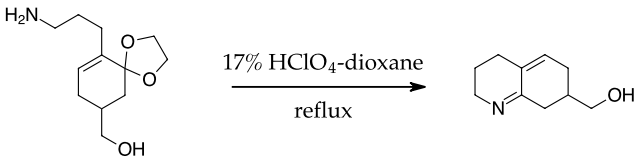
"Effective Combination of Two-Directional Synthesis and Rhenium(VII) Chemistry: Total Synthesis of meso Polyether Teurilene." Y. Morimoto et al., *J. Am. Chem. Soc.*, **1999**, *121*, 6792-6797. https://doi.org/10.1021/ja990154i

6.



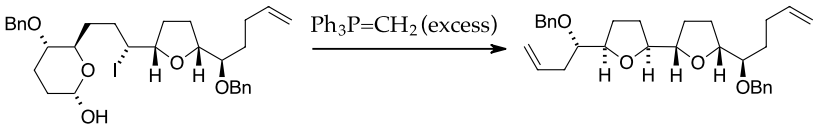
"Total syntheses of (+)-ipomeamarone and (−)-ngaione via novel intramolecular hydrogen abstraction." T. Sugimura et al., *Tetrahedron*, **1994**, *50*, 11647-11658. https://doi.org/10.1016/S0040-4020(01)85659-3

7.



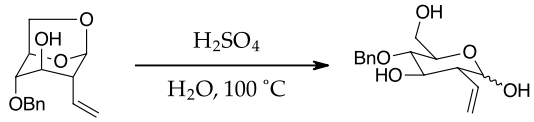
"The First Total Synthesis of (±)-Huperzine B." B. Wu, D. Bai, *J. Org. Chem.*, **1997**, *62*, 5978-5981. https://doi.org/10.1021/jo970248f

8.



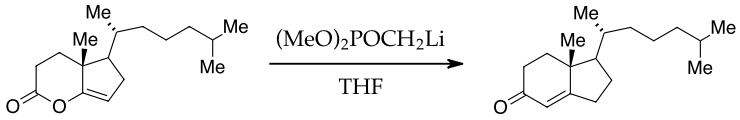
"A Modular Synthesis of the Bis-Tetrahydrofuran Core of Rolliniastatin from Pyranoside Precursors." Z. Ruan at al., *Tetrahedron*, **2000**, *56*, 9203-9211. https://doi.org/10.1016/S0040-4020(00)00878-4

9.



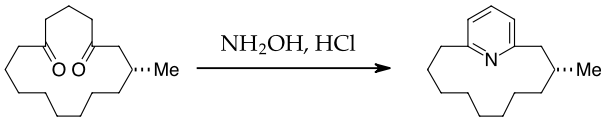
"Synthesis of the first pseudosugar-C-disaccharide. A potential antigen for eliciting glycoside-bond forming antibodies with catalytic groups." C. Barbaud et al.,, *Tetrahedron*, **1995**, *51*, 9063-9078. https://doi.org/10.1016/0040-4020(95)00505-3

10.



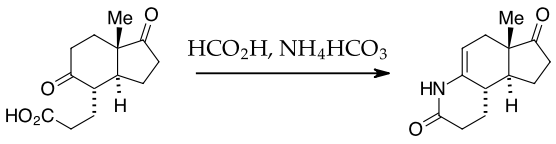
"Synthesis of a Vitamin D_3 Hydrindan Ring–Side-Chain Building Block, Involving Tandem Conjugate Addition and Alkylation Reactions." P. Grzywacz et al., *J. Org. Chem.*, **1997**, *62*, 5293-5298. https://doi.org/10.1021/jo9705834

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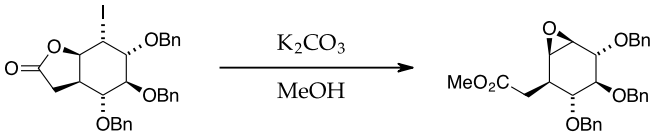
"Application of Ring Closing Metathesis to the First Total Synthesis of (R)-(+)-Muscopyridine: Determination of Absolute Stereochemistry." H. Hagiwara et al., *J. Org. Chem.*, **2000**, *65*, 7231-7234. https://doi.org/10.1021/jo000785r

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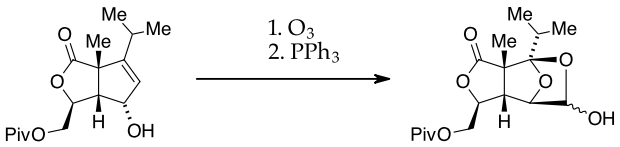
"A Concise Route to 19-Nor-10-azasteroids, a New Class of Steroid 5α-Reductase Inhibitors. 3. Synthesis of (+)-19-Nor-10-azatestosterone and (+)-17β-(Acetyloxy)-(5β)-10-azaestr-1-en-3-one." A. Guarna et al., *J. Org. Chem.*, **1998**, *63*, 4111-4115. https://doi.org/10.1021/jo971917s

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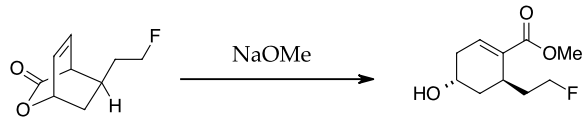
"Free Radical Studies and Solutions to the Synthesis of (+)-Cyclophellitol." F.E. Ziegler, Y. Wang, *J. Org. Chem.*, **1998**, *63*, 7920-7930. https://doi.org/10.1021/jo981211d

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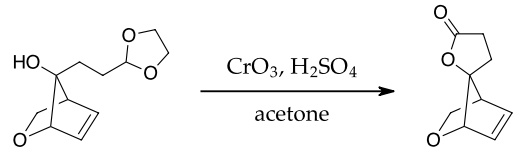
"Novel Total Synthesis of (+)-Eremantholide A." Ken-ichi Takao et al., *J. Org. Chem.*, **1995**, *60*, 8179-8193. https://doi.org/10.1021/jo00130a017

15.



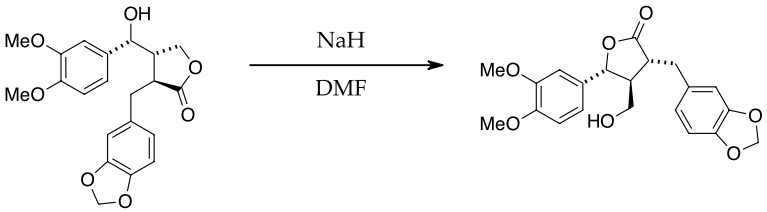
"Antiproliferative Hybrid Analogs of the Hormone 1α,25-Dihydroxyvitamin D_3: Design, Synthesis, and Preliminary Biological Evaluation?" G.H. Posner et al., *J. Org. Chem.*, **1997**, *62*, 3299-3314. https://doi.org/10.1021/jo970049w

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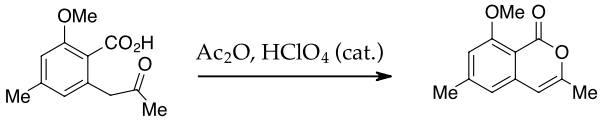
"Enantioselective Synthesis of (−)-Curcumanolide A Using Enzymatic Transesterification of meso-Spirodiol." T. Fujita et al., *J. Org. Chem.*, **1997**, *62*, 3824-3830. https://doi.org/10.1021/jo962150r

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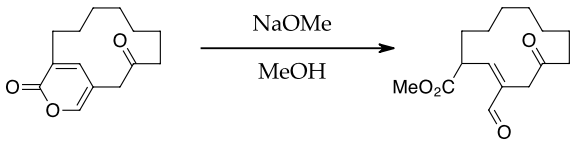
"Stereoselective Syntheses of Cis- and Trans-Isomers of α-Hydroxy-α,β-dibenzyl-γ-butyrolactone Lignans: New Syntheses of (±)-Trachelogenin and (±)-Guayadequiol." Y. Moritani et al., *J. Org. Chem.*, **1996**, *61*, 6922-6930. https://doi.org/10.1021/jo9601932

18.



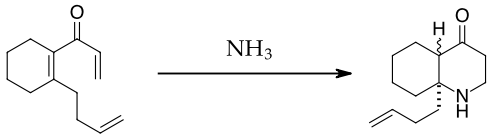
"Total synthesis of novel xanthone antibiotics (±)-cervinomycins A1, and A2." G. Mehta et al., *Tetrahedron*, **1994**, *50*, 11729-11742. https://doi.org/10.1016/S0040-4020(01)85666-0

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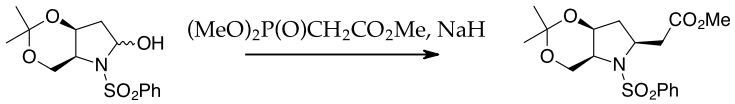
"Flexible Synthesis of Metacycloprodigiosin and Functional Derivatives Thereof." A. Fürstner, H. Krause, *J. Org. Chem.*, **1999**, *64*, 8281-8286. https://doi.org/10.1021/jo991022a

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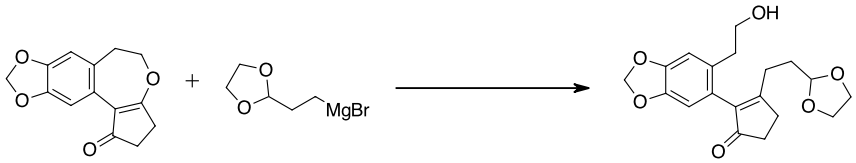
"Total Synthesis of (±)-Cylindricines A and B." J.F. Liu, C.H. Heathcock, *J. Org. Chem.*, **1999**, *64*, 8263-8266. https://doi.org/10.1021/jo991020q

21.



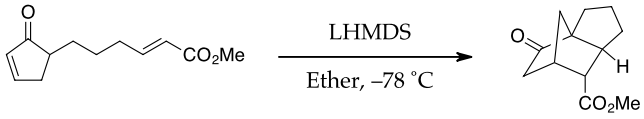
"Conformationally Constrained 7-Azabicyclo[2.2.1]heptane Amino Acids. Synthesis of a Glutamic Acid Analogue." B.P. Hart, H. Rapoport, *J. Org. Chem.*, **1999**, *64*, 2050-2056. https://doi.org/10.1021/jo982327c

22.



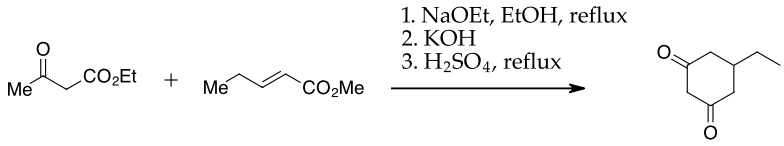
"Two Interrelated Strategies for Cephalotaxine Synthesis." X. Lin et al., *J. Org. Chem.*, **1996**, *61*, 7335-7347. https://doi.org/10.1021/jo961179s

23.



"Facile Construction of the Tricyclo[5.2.1.0^{1,5}]decane Ring System by Intramolecular Double Michael Reaction: Highly Stereocontrolled Total Synthesis of (±)-8,14-Cedranediol and (±)-8,14-Cedranoxide." M. Ihara et al., *J. Org. Chem.*, **1999**, *64*, 1259-1264. https://doi.org/10.1021/jo981996n

24.



"Total Synthesis of Angucyclines. Part 15: A Short Synthesis of (±)-6-Deoxybrasiliquinone B." K. Krohn et al., *Tetrahedron*, **2000**, *56*, 4753-4758, https://doi.org/10.1016/S0040-4020(00)00398-7

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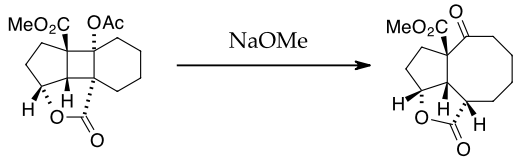
"Asymmetric Syn-Selective Aldol Reactions of γ-Oxygenated Vinylogous Urethane with a Second Generation Chiral Auxiliary: Application in Construction of (+)-3-Deoxy-D-manno-2-octulosonic Acid." R.H. Schlessinger, L.H. Pettus, *J. Org. Chem.*, **1998**, *63*, 9089-9094. https://doi.org/10.1021/jo981916f

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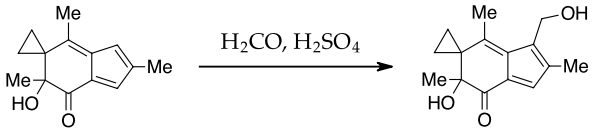
"Optical Resolution of 3-(Silyloxy)glutaric Acid Half Esters and Their Utilization for Enantioconvergent Synthesis of a HMG-CoA Reductase Inhibitor." T. Konoike et al., *J. Org. Chem.*, **1998**, *63*, 3037-3040. https://doi.org/10.1021/jo9722156

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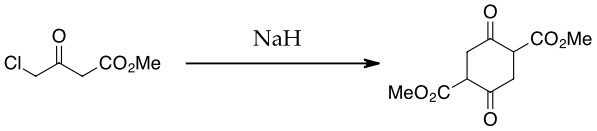
"Use of Cyclic α-Keto Ester Derivatives in Photoadditions. Synthesis of (±)-Norasteriscanolide." G.L. Lange, M.G. Organ, *J. Org. Chem.*, **1996**, *61*, 5358-5361. https://doi.org/10.1021/jo960088s

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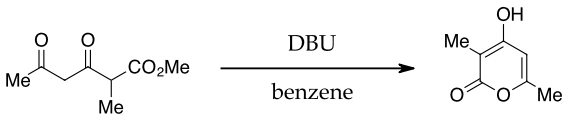
"A Rapid Synthesis of Hydroxymethylacylfulvene (HMAF) Using the Allenic Pauson-Khand Reaction. A Synthetic Approach to Either Enantiomer of This Illudane Structure." K.M. Brummond et al., *J. Am. Chem. Soc.*, **2000**, *122*, 4915-4920. https://doi.org/10.1021/ja000546l

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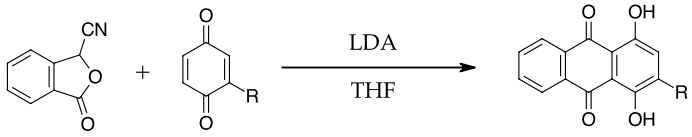
"Asymmetric Heck Reaction?Carbanion Capture Process. Catalytic Asymmetric Total Synthesis of (−)-$\Delta^{9(12)}$-Capnellene." T. Ohshima et al., *J. Am. Chem. Soc.*, **1996**, *118*, 7108-7116. https://doi.org/10.1021/ja9609359

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"Total Synthesis of (−)-Solanapyrone A via Enzymatic Diels-Alder Reaction of Prosolanapyrone." H. Oikawa et al., *J. Org. Chem.*, **1998**, *63*, 8748-8756 https://doi.org/10.1021/jo980743r

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"The synthesis of bisanthraquinone derivatives as DNA bisintercalating agents." P. Ge, R.A. Russell, *Tetrahedron*, **1997**, *53*, 17477-17488. https://doi.org/10.1016/S0040-4020(97)10196-X

Learning Goals

Through the suggested textbook reading, problems and participating in this class discussion, students will have…

- reviewed more organic chemistry mechanisms and are able to describe reactions that have been seen again and again.
- developed a vocabulary for chemical mechanism and are able to describe mechanism using words as well as with diagrams.